

Expedition to the Bay of Bengal

By Alysha Blömker, 30.03.26

It was the end of dry season, lining the roads of Colombo were piles of golden yellow mangos and street vendors meandered the still standing traffic to convince you that these are the best fruits you will ever



Figure 1 The Anna Weber van Bosse



Figure 2 Mangos on the streets of Colombo

taste. They were right! Marloes, Jesse and I had flown 13 hours and 35 minutes with huge pieces of luggage later rattling in the back of our taxi. In addition to the normal three-week luggage, we packed safety equipment (e.g. helmets, protective clothing) and some personal food that wouldn't be available on the ship (lots of gummy bears!). In that moment our goal was to reach the port of Colombo in time; our research vessel the Anna Weber van Bosse was waiting for us. Seeing her anchored at the shore I thought "this is it – here we'll live for the next 17 days" and I felt a surge of excitement and anticipation.

The first rule of planning an expedition is "nothing goes as planned" ...

Research Aim: Monitoring the effect of monsoon rainfall on water column salinity stratification

It comes to no surprise that there is salt in the ocean - what is interesting is how much salt there is at different depths; This is what we refer to as salinity stratification. At the bottom is the cold salty water and at the top is the warmer fresh water. The amount of surface fresh water is determined by a balance of precipitation and evaporation. Increasing precipitation decreases salt concentration, whilst increasing evaporation increases salt concentration – they are two mechanisms working in opposition to each other. During monsoon season there is a strong increase in precipitation – it can rain heavily for days on end. At the same time the sky is overcast with brooding clouds, thus evaporation decreases. All this together led us to hypothesize a larger layer of fresh water sitting at the top of the water column.

Our method of quantifying this layer is to calculate the mixed layer depth. The mixed layer is characterized by nearly uniform stratification due to higher turbulence (e.g. from waves and wind at the surface) and the point at which this region transitions to the more stratified lower part of the column is the mixed layer depth [1].

Argos Floats are small probes which float along the ocean current and can dive up and down to capture a vertical profile. They can independently stay in the ocean for multiple weeks.

CTDs (Conductivity, Temperature and Depth) are made up of large clusters of sensors that are lowered from a research ship. They produce a vertical profile of conductivity (to infer salinity) and temperature.



Figure 3 Jesse deploying our 1st Argo!

The engines started, the anchors were drawn and we stood on deck, under the intense mid-day sun, and watched Colombo fade into the horizon. Next – move to the first way point of our expedition! We planned to cover 15 waypoints over the course of the 17 days, half on our way to the Bangladeshi coast, the other half on the way back. This was my first time on the open ocean and to fight sea sickness I had packed dimenhydrinate to supply the entire ship, but that was not sadly enough... Whilst Marloes was able to join

the first days of CTD and Argo deployment, Jesse and I were sitting in our cabins feeling like we had the worst hangover of our lives. Thankfully the sea was rather calm (it was just before the start of monsoon season after all) and we grew our pair of sea legs within a few days!

We were incredibly lucky with our crew as well. Thijmen and Hilda are both lovely people and great technicians. Hilda literally showed us the ropes as she explained the ins and outs of the CTD hoisting mechanism and on day 7 we (unfortunately) got to see her mechanical expertise in action. I was just about to get coffee when Jesse ran up to me “The CTD is stuck! Hilda and Thijmen are trying to fix it!” and I was so worried – we had planned quite a tight schedule, if we can’t make it to the next waypoint today, we’ll lose out on a deployment of an Argo. As I nervously sat in my cabin inspecting the salinity profiles that we gather from the CTD deployments, Hilda and Thijmen were able to repair the coiling mechanism within 5 hours. Our journey continued (with a small delay)!



Figure 4 Thijmen bringing the CTD back on board.

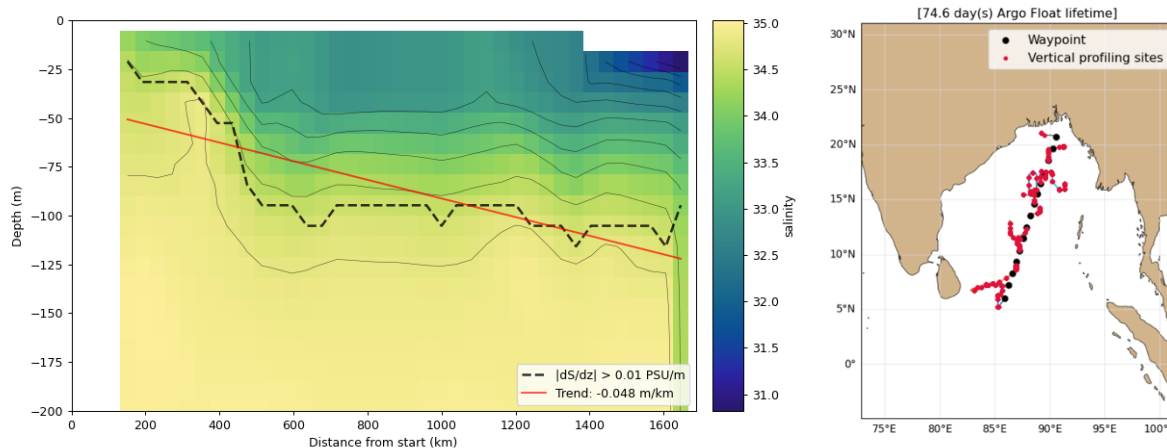


Figure 5 Left: An example of the salinity profiles from the Argo Floats. The red line shows the mixed layer depth, whilst the dashed line shows a threshold of the salinity gradient. Right: Map of the Argo measuring points. We started the expedition at the most southern point and sailed northeast towards the coast.

Fig. 5 shows what a salinity profile measured by our Argo Floats looks like. As we move towards the coast, there is more fresh water at the surface (due to river outflow) and thus a deeper mixed layer. The next step is to link these profiles to precipitation measurements to see whether high amounts of rain at those locations change the mixed layer depth!

As we approached the Bangladeshi coast the weather started to change. First, the intense sun was diffused by clouds, then it started to rain lightly. We had quite high hopes that we would be able to witness enough precipitation events over the next two months with the Argo floats. They remain in the water, even after we are back in the Netherlands.

The Importance of Salinity Stratification

Studying mixed layer depth in the Bay of Bengal is specifically interesting as it can be used to improve predictions of tropical cyclones [2]. In the past two decades stronger salinity stratification in the northern part of the Bay of Bengal have been linked to faster growing and more intense tropical cyclones [3]. It is thought that the deeper mixed layer prevents tropical cyclones from cooling the sea surface, keeping the cyclone alive longer [3].

The return line went smoothly; we hit all way points and despite minor hiccups in communication with the Argos, were able to deploy all we intended to. We had found our routine on the ship, and I especially enjoyed our little afternoon tea ritual—our captain Gayan brought Ceylon tea from his home region. It was packaged in a metal tin ornamented with golden Bengal tigers, the symbol of unlimited power.

After leaving our final waypoint we once again stood on the deck and scanned the horizon. And there they were - the concrete buildings of Colombo emerging from the sea. Land after all!

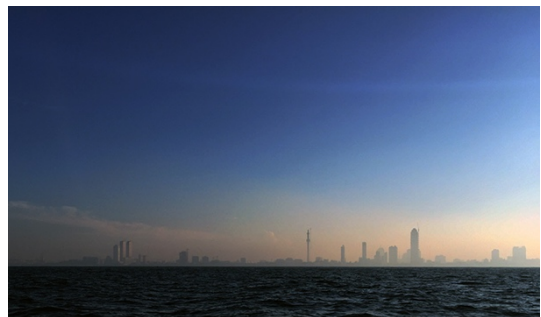


Figure 6 The skyline of Colombo.

Text References

- [1] https://journals.ametsoc.org/view/journals/atot/20/2/1520-0426_2003_020_0319_emldfo_2_0_co_2.xml
- [2] <https://rmets.onlinelibrary.wiley.com/doi/full/10.1002/met.1862>
- [3] <https://iopscience.iop.org/article/10.1088/1748-9326/ab9690>

Image References

- Figure 1: Alysha Blömker
- Figure 2: <https://www.whoi.edu/what-we-do/explore/instruments/instruments-sensors-samplers/conductivity-temperature-depth-ctd-sensors/>
- Figure 3: <https://www.aoml.noaa.gov/argo-the-crown-jewel-of-ocean-observing-systems-turns-25/>
- Figure 4: <https://timeline.nioz.nl/AnnaWeber/>
- Figure 5: <https://divesrilanka.com/SCUBA%20diving%20in%20Colombo%20Sri%20Lanka.html>

AI Disclosure Statement

No AI was used before or during the writing of this text.
No AI was used (by me) to produce or modify the images.